

Daniel Medeiros

Solutions Architect & Applied AI/HPC Researcher

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- **Location:** Stockholm, Sweden

Professional Summary

Solutions-oriented researcher and engineer with 7+ years of experience across HPC, cloud-native infrastructure, and applied AI. PhD from KTH Royal Institute of Technology specializing in scaling HPC workloads on cloud systems. Currently helping SMEs and public-sector organizations adopt large-scale AI through EuroHPC AI Factory initiatives at RISE Research Institutes of Sweden — designing proof-of-concepts, delivering technical demos, and onboarding customers onto GPU clusters. Proven ability to bridge deep technical expertise with customer-facing engagement, translating complex architectures into actionable solutions. Background spans GPU programming (CUDA, HIP, OpenCL), cluster management, containerized workflows, and end-to-end AI pipeline development on multi-node GPU systems.

Core Competencies

Domain	Technologies & Skills
HPC & Cluster Management	Slurm, MPI, OpenMP, job scheduling, multi-node GPU workloads
GPU & Accelerated Computing	CUDA, HIP, OpenCL, multi-GPU distribution, profiling
Cloud & Orchestration	Kubernetes, Docker/Podman, Helm, Ceph Object Storage, Prometheus
AI/ML Frameworks	PyTorch, scikit-learn, OpenCV, Diffusers, MLflow, vLLM
Programming	Python, C/C++, Bash, Linux systems administration
DevOps & CI/CD	Jenkins, Git, container registries, infrastructure-as-code
Customer Engagement	Technical demos, proof-of-concepts, pre-sales support, CRM (Pipedrive), trainings/workshop delivery

Professional Experience

Researcher — RISE Research Institutes of Sweden

2025 – Present | Stockholm, Sweden

- **AI Factory Solutions Architecture:** Design and deliver end-to-end AI proof-of-concepts for SMEs and public-sector organizations under the Swedish AI Factory (MIMER) and ENCCS projects, funded by EuroHPC-JU together with the Swedish state.
- **GPU Cluster Operations:** Deploy and optimize AI inference and generation pipelines on Leonardo supercomputer infrastructure (NVIDIA A100 GPUs), including multi-GPU distribution, memory management, and containerized workloads via Slurm and Kubernetes.
- **Generative AI Demos:** Architected and delivered interactive AI demos for conference booth presentations, including real-time video generation (LTX-Video), face-swapping pipelines, and voice cloning systems — managing the full stack from model selection to GPU resource allocation.
- **Customer & Partner Engagement:** Manage a lead pipeline via CRM, drive collaborations through startup incubators and AI events, and provide hands-on technical guidance to customers adopting HPC and AI infrastructure.
- **Training & Enablement:** Organize and deliver technical workshops on GPU Programming, Cloud Computing, Containerization, and Distributed Systems for industry and academic audiences.

Geophysicist — Geological Survey of Brazil (SGB/CPRM)

2014 – 2021 | Rio de Janeiro, Brazil

- **Automation & Software Development:** Built Python-based tools and plugins for ArcGIS and Oasis montaj (GDAL, GeoPandas), automating geophysical data workflows and reducing manual processing time.
- **Data Processing:** Managed acquisition, QC, and processing of aerogeophysical datasets (gravimetric, magnetic, spectrometric) at national survey scale.
- **Scientific Output:** Co-authored book chapters, technical reports, and regional geological maps published by the Brazilian geological survey.

Education

PhD in Computer Science — KTH Royal Institute of Technology

2021 – 2025 | Stockholm, Sweden

- **Thesis:** "Towards Adaptive Resource Management for HPC Workloads in Cloud Environments"
- **EU Research Projects:** Coordinated 3 Work Package deliverables and co-authored 1 for Horizon Europe projects (OpenCUBE, IO-SEA), involving multi-partner collaboration across European HPC centers.
- **Key research areas:** Cloud-native HPC, container orchestration on GPU clusters, performance benchmarking, and scheduling optimization.

MSc in Computer Science — Universidade Federal da Bahia

2019 – 2021 | Salvador, Brazil

- **Thesis:** "Intelligent scheduling of search queries using Elasticsearch on DVFS-enabled processors".
- **Relevant coursework:** Real-Time Systems, Machine Learning, Applied Statistics.

BSc in Geophysics — Universidade Federal do Rio Grande do Norte

2010 – 2013 | Natal, Brazil

- Geophysics of Petroleum Exploration, Seismic Data Processing, Reservoir Geophysics.

Selected Projects & Technical Highlights

- **Real-Time AI Video Generation Pipeline:** Designed an interactive system using LTX-Video models for image to video generation at conference demos, running on distributed A100 GPU nodes.
- **Multi-Framework GPU Portability:** Research and benchmarking across CUDA, HIP, OpenCL, and SYCL for performance-portable GPU code, contributing to open-source HPC tooling.
- **Container-Native HPC Workflows:** Developed and deployed containerized AI/HPC applications on Kubernetes and Slurm-managed clusters, including GPU-aware scheduling and resource isolation.
- **HPC Onboarding for Enterprise:** Guided multiple organizations through their first deployments on EuroHPC supercomputers (LUMI, Leonardo), and helped them design their own solutions for these environments.

Leadership & Community

- **KTH University Board Member (2022–2024):** Represented 1,500+ doctoral students on KTH's highest governing body (*Universitetsstyrelsen*). Served as Chairperson, Auditor, and Webmaster of the PhD Chapter; managed a 1M SEK annual budget and a team of 10+ representatives.
- **Sveriges förenade studentkårer - Doktorandkommittén (2024–2025):** Elected member to the committee that represents over 17 000 doctoral students in the country, directly engaging in related important questions for the community such as migration aspects, and the third-level course database.
- **Action10.org (2025):** Webmaster, redesigned the organization's website to improve donor and audience engagement.

Languages

- **English:** Fluent
- **Portuguese:** Native
- **Swedish:** Professional working proficiency

Publications

A complete list of publications in Computer Science is available on [DBLP](#).